Question #1 of 77

Question ID: 1473654

At the inception of a market-rate plain vanilla swap, the value of the swap to the fixed-rate payer is:

A) positive.

B) zero.

C) either positive or negative.

Explanation

A market-rate swap is priced so that the value to either side is zero at the inception of the swap.

(Module 30.6, LOS 30.e)

Chantal DuPont is the CFO of Vetements Verdun, a manufacturer of specialty clothing and uniforms, located in northern France. The firm is currently undergoing an expansion which will require DuPont to draw down 25 million on Vetements Verdun's credit line as a 90-day bridge loan before the mortgage closes. The money will not be needed for 60 days, at which point the interest rate will be determined. The interest rate on the loan will be based off 90-day LIBOR.

DuPont is becoming concerned because of signs that interest rates may begin to rise. The firm cannot afford to have its borrowing costs increase significantly over current rates. In response to DuPont's concerns, the company's CEO, Viviane Lamarre, has asked DuPont to hedge the firm's borrowing costs, even if that entails some near-term outlays.

DuPont and Lamarre discuss entering into a forward rate agreement (FRA) to hedge Vetements Verdun's interest rate exposure on the credit line. Current LIBOR rates are:

LIBOR rate	
30-day	2.6%
60-day	2.8%
90-day	3.0%
120-day	3.2%
150-day	3.3%

180-day	3.4%

They decide to go forward with the hedge and DuPont enters into the appropriate FRA for the full amount of 25 million.

In the first 30 days of the FRA, the fixed income markets rally sharply. The new set of LIBOR rates, on the thirtieth day of the FRA, is:

LIBOR rate	
30-day	2.2%
60-day	2.4%
90-day	3.6%
120-day	3.8%
150-day	3.8%
180-day	3.8%

At the settlement date, the interest savings on the loan term is 23,750. DuPont tells Lamarre, "I am looking forward to cashing our settlement check for 23,750." Lamarre adds, "Yes, and on top of that we get to borrow for 90 days at a below-market rate." Both DuPont and Lamarre are pleased with their decision to hedge.

Question #2 - 5 of 77

Question ID: 1586277

Which statement *most* accurately describes a 2 x 3 forward rate agreement?

A) Contract expires in two months on an underlying loan settled in three months.

B) Two-month underlying interest rate on a contract settled in three months.

C) Underlying loan of two month maturity under a contract that expires in three months.

Explanation

A 2 x 3 forward rate agreement is a contract that expires in two months and the underlying loan is settled in three months. The underlying rate is a 30-day (1-month) rate on a 30-day (1-month) loan in 60 days (2 months).

(Module 30.4, LOS 30.c)

Question #3 - 5 of 77

Question ID: 1586278

Which forward rate agreement would *most* effectively hedge Vetements Verdun's exposure to LIBOR?

A) 3 x 2.

B) 2 x 5.

C) 2 x 3.

Explanation

Vetements Verdun needs to be hedged against 90-day LIBOR rates that will prevail 60 days from now. Such a hedge would require a two-month contract on three-month rates, to be settled in five months: a 2 x 5.

(Module 30.4, LOS 30.c)

Question #4 - 5 of 77

Question ID: 1586279

Which value is *closest* to the price of the most effective hedge for Vetements Verdun?

A) 3.3%.

B) 3.6%.

C) 3.0%.

Explanation

The actual, unannualized rate on the 60-day loan is:

$$R60 = 0.028 \times 60/360 = 0.00467$$

The actual, unannualized rate on the 150-day loan is:

$$R150 = 0.033 \times 150/360 = 0.01375$$

So the rate on a 90-day loan to be made 60 days from now is:

$$FR\ (60,90) = ((1 + R150)/(1 + R60)) - 1$$

$$FR\ (60,90) = (1.01375/1.00467) - 1$$

$$FR\ (60,90) = 1.00904 - 1$$

$$FR\ (60,90) = 0.904\%$$

We annualize this rate using the formula:

$$0.904\% \times (360/90) = 3.62\%$$

(Module 30.4, LOS 30.c)

Typesetting math: 100%

Question #5 - 5 of 77

Question ID: 1586280

What must the 90-day LIBOR rate have been at the expiration of the contract?

A) 3.6%.

B) 4.0%.

C) 3.4%.

Explanation

Since Vetements Verdun is long the FRA, the market rate of interest at settlement must be higher than the price of the contract and the 23,750 has a positive value. The interest savings at the end of the loan term will be:

Interest savings = ((market rate × (90/360)) − (0.0362 × (90/360))) × 25,000,000

23,750 = ((market rate × 90/360) − 0.00905) × 25,000,000

0.000950 = market rate × 90/360 − 0.00905

0.0100 = market rate × 0.25

0.0400 = market rate

The market rate must have been 4.0%.

(Module 30.4, LOS 30.c)

Question #6 of 77

Question ID: 1473669

The current U.S. dollar ($) to Canadian dollar (C$) exchange rate is 0.7. In a $1 million currency swap, the party that is entering the swap to hedge existing exposure to C$-denominated fixed-rate liability will:

A) pay C$1,428,571 at the beginning of the swap.

B) pay floating in C$.

C) receive floating in C$.

Explanation

The party that is entering the swap to hedge existing exposure to C$-denominated fixed-rate liability will want to receive-fixed C$. They will pay 1,000,000/0.7 = C$1,428,571 (principal) at swap inception (in exchange for USD 1 million) and get the same amount (C$1,428,571) back at termination (in exchange for paying back the USD 1 million).

 , LOS 30.f)

Question #7 of 77

Question ID: 1473641

Calculate the price (expressed as an annualized rate) of a 1x4 forward rate agreement (FRA) if the current 30-day rate is 5% and the 120-day rate is 7%.

A) 7.47%.

B) 6.86%.

C) 7.63%.

Explanation

A 1x4 FRA is a 90-day loan, 30 days from today.

The actual rate on the 30-day loan is: R30 = 0.05 x 30/360 = 0.004167

The actual rate on the 120-day loan is: R120 = 0.07 x 120/360 = 0.02333

FR (30,90) = [(1+ R120)/(1+ R30)] – 1 = (1.023333/1.004167) – 1 = 0.0190871

The annualized 90-day rate = 0.0190871 x 360/90 = .07634 = 7.63%

(Module 30.4, LOS 30.c)

Question #8 of 77

Question ID: 1473672

Consider a one-year currency swap with semiannual payments. The payments are in U.S. dollars and euros. The current exchange rate of the euro is $1.30 and interest rates are

	180 days	360 days
USD MRR	5.6%	6.0%
MRR	4.8%	5.4%

What is the fixed rate in euros?

A) 5.318%.

B) 2.659%.

C) 5.245%.

Explanation

Typesetting math: 100%

The present values of 1 euro received in 180 days and 1 euro received in 360 days are:

$$1/(1 + 0.048 \times (180/360)) = 0.9766 \text{ and } 1/1.054 = 0.9488$$

The fixed rate in euros is (1 - 0.9488) / (0.9766 + 0.9488) = 0.026592 × (360/180) = 5.318%. The notional principal is 100,000/1.30 = 76,923 euros.

(Module 30.7, LOS 30.f)

Question #9 of 77

Question ID: 1473617

The contract price of a forward contract is:

A) always the present value of the expected future spot price.

B) determined at the settlement date.

C) the price that makes the contract a zero-value investment at initiation.

Explanation

The contract price can be an interest rate, discount, yield to maturity, or exchange rate. The forward price is the future value of the spot price adjusted for any periodic payments expected from the asset. An example of when the forward price may be less than the spot price is in the case of an equity index contract where the dividend yield is greater than the risk-free rate.

(Module 30.1, LOS 30.b)

Question #10 of 77

Question ID: 1473662

Writing a series of interest-rate puts and buying a series of interest-rate calls, all at the same exercise rate, is equivalent to:

A) being the floating-rate payer in an interest rate swap.

B) being the fixed-rate payer in an interest rate swap.

C) a short position in a series of forward rate agreements.

Explanation

A short position in interest rate puts will have a negative payoff when rates are below the exercise rate; the calls will have positive payoffs when rates exceed the exercise rate. This mirrors the payoffs of the fixed-rate payer who will receive positive net payments when settlement rates are above the fixed rate.

, LOS 30.e)

Question #11 of 77

Question ID: 1473625

The no-arbitrage price of a futures contract with a spot rate of 990, a time to maturity of 2 years, and a risk-free-rate of 5% is *closest* to:

A) 1091

B) 1040

C) 792

Explanation

The no-arbitrage price of a futures contract is based on the spot rate, the time to maturity, and the risk-free-rate.

FP	$= S_0 \times (1 + R_f)^T$
	$= 990(1.05)^2$
	$= 1091$

(Module 30.1, LOS 30.b)

Question #12 of 77

Question ID: 1473655

The fixed-rate on a semiannual 2-year interest rate swap is *closest* to the:

A) coupon rate on a 2-year par bond with the same credit risk as the reference rate.

B) coupon rate on a 2-year par bond with the same credit risk as the fixed-rate payer.

C) current 180-day T-bill rate.

Explanation

The fixed-rate on a swap is calculated using the yield curve for the floating rate reference. Therefore, the fixed rate reflects the credit spread of that rate over the riskless rate of return.

(Module 30.6, LOS 30.e)

The Isle of Nefer is a developing country with its stock and futures markets enjoying record trading volumes due to the influx of foreign funds. You are looking to invest in the stock and futures markets in the Isle of Nefer. The representative stock market index, Nefer Industrial Index (NII), is currently priced at 8,765 and the one year NII future contract is currently trading at 8,920.

You have experience in using forward contracts but not futures. You discuss the possibility of investing in the Isle of Nefer using futures contract with your supervisor, Peter Filler, and he makes the following comments.

Comment 1:	"A futures contract will have positive value after marking to market if the future price is up on that day."
Comment 2:	"Given a quoted clean bond price of the CTD, when looking at a bond future the full price of the bond must be used which equals the clean price of the bond plus accrued interest. The futures price can then be calculated as: QFP = {(full price) × $(1 + R_f)^T$ + AI_T – FVC)(1 / CF) Where AI_T = accrued interest at futures maturity, R_f = risk-free rate, FVC = future value of coupon and CF = conversion factor

Peter Filler also suggests that you invest in Treasury bond futures. Exhibit 1 contains the relevant information.

Exhibit 1

Price of underlying deliverable 16 year 5% Treasury bond (just paid coupon)	$1,030
Expiration of Treasury bond futures contract	0.7 year
Conversion factor	1.08
Risk free rate	3.0 percent

Question #13 - 16 of 77

Question ID: 1473637

How will the price of the one-year stock index future perform over the next 12 months?

Typesetting math: 100%

A) The futures price will converge to the future spot index price, with the basis reducing to zero.

B) The value will move approximately in line with the spot index price, with a fairly constant basis.

C) The futures price will move approximately in line with the spot index price, though its actual level at the end of the year will depend more on supply and demand than on the spot price.

Explanation

The futures price will converge to the spot price over the lifetime of the contract, with the basis (difference between future and spot) reducing over this period to zero as an expiring futures contract is the same as a spot transaction. At expiration the futures and spot price will converge to prevent an arbitrage opportunity. Answer B is wrong, as it implies the future price will move in a straight line.

(Module 30.3, LOS 30.d)

Question #14 - 16 of 77

Question ID: 1473638

Comment 1 is *best* described as:

A) correct.

B) incorrect as the value of the futures contract should be negative after marking to market.

C) incorrect as the value of the futures contract should be zero after marking to market.

Explanation

The value of a futures contract will reset to zero after marking-to-market.

(Module 30.3, LOS 30.d)

Question #15 - 16 of 77

Question ID: 1473639

Comment 2 is *best* described as:

A) correct.

B) incorrect as the full price should be the clean price less the current accrued interest.

C) incorrect as the accrued interest at expiration should be deducted from the future value of full bond price in arriving at the quoted futures price.

Explanation

The quoted (clean) price of the futures contract is the future value of the full price of the cheapest to deliver bond less accrued interest (since last coupon) and the future value any coupon payment received during the life of the contract and then adjusted for the appropriate conversion factor.

QFP = {(full price) × $(1 + R_f)^T$ – AI_T – FVC)(1 / CF)

(Module 30.3, LOS 30.d)

Question #16 - 16 of 77

Question ID: 1473640

Using Exhibit 1, the quoted price of the Treasury bond futures contract should be *closest* to:

A) $941.

B) $975.

C) $1,100.

Explanation

An investor of the Treasury bond will receive one semi-annual coupon in 0.5 years from now (or 0.2 years before maturity). At expiration of the futures contract, the CTD bond will have 0.2 years of accrued interest.

FV of coupon (FVC) = $25 × $1.03^{(0.7 – 0.5)}$ = $25.15

AI_T = 0.2 / 0.5 × $25 = $10

QFP = {(full price) × $(1 + R_f)^T$ – AI_T – FVC)(1 / CF)

= [1030(1.03)^{0.7}$ – 10 – 25.15](1 / 1.08)

= $941.10

(Module 30.3, LOS 30.d)

Typesetting math: 100%

Question #17 of 77

Question ID: 1473673

90 days ago the exchange rate was USD 0.83 per CDN and the term structure was:

	180 days	360 days
USD MRR	5.2%	5.6%
CDN MRR	4.8%	5.4%.

A 1 year, semi-annual settlement, fixed for fixed swap was initiated with 5.30% fixed for CDN and 5.52% fixed for USD on a principal of USD 1 million.

Current exchange rate is USD 0.84 per CDN and the yield curve is:

	90 days	270 days
USD MRR	5.2%	5.6%
Disc Factor	0.98717	0.95969
CDN MRR	4.8%	5.4%
Disc factor	0.98814	0.96108

What is the value of the swap to the USD interest payer?

A) $11,500.

B) $10,126.

C) -$3,472.

Explanation

CDN principal = 1,000,000/0.83 = CDN 1,204,819. CDN payments = (0.053/2) x 1,204,819 = CDN 31,928.

USD payment = 1,000,000 x (0.0552/2) = USD 27,600

Value of CDN payments = 31,928 x 0.98814 + (1,204,819 + 31,928) x 0.96108 = CDN 1220162

Value in USD = 1220162 x 0.84 = USD 1,024,936.

Value of USD payments = 27,600 x 0.98717 + 1,027,600 x 0.95969 = USD 1,013,423

Value to USD interest payer = $1,024,936 – $1,013,423 = $11,513

(Module 30.7, LOS 30.f)

Typesetting math: 100%

Question #18 of 77

Question ID: 1473627

To initiate an arbitrage trade if the futures contract is underpriced, the trader should:

A) borrow at the risk-free rate, short the asset, and sell the futures.

B) short the asset, invest at the risk-free rate, and buy the futures.

C) borrow at the risk-free rate, buy the asset, and sell the futures.

Explanation

If the futures price is too low relative to the no-arbitrage price, buy futures, short the asset, and invest the proceeds at the risk-free rate until contract expiration. Take delivery of the asset at the futures price, pay for it with the loan proceeds and keep the profit. For Treasury bill (T-bills), shorting the asset is equivalent to borrowing at the T-bill rate.

(Module 30.1, LOS 30.b)

Question #19 of 77

Question ID: 1473665

If the one year spot rate is 5%, the two-year spot rate is 5.5%, and the three year spot rate is 6%, the fixed rate on a 3-year annual pay swap is *closest* to:

A) 1.99%.

B) 4.50%.

C) 5.65%.

Explanation

The fixed rate on the swap is: $\frac{1-\frac{1}{1+0.06(3)}}{\frac{1}{1.05}+\frac{1}{1+0.055(2)}+\frac{1}{1+0.06(3)}}$

$$\frac{1-0.8475}{0.9524+0.9009+0.8475}$$

= 0.1525 / 2.7008 = 0.0565

(Module 30.6, LOS 30.e)

Question #20 of 77

Question ID: 1586281

Typesetting math: 100%

30 days ago, J. Klein took a short position in a $10 million (3X6) forward rate agreement (FRA) based on MRR and priced at 5%. The current MRR curve is:

- 30-day = 4.8%
- 60-day = 5.0%
- 90-day = 5.1%
- 120-day = 5.2%
- 150-day = 5.4%

The current value of the FRA, to the short, is *closest* to:

A) −$15,154.

B) −$15,495.

C) −$15,280.

Explanation

FRAs are entered in to hedge against interest rate risk. A person would buy a FRA anticipating an increase in interest rates. If interest rates increase more than the rate agreed upon in the FRA (5% in this case) then the long position is owed a payment from the short position.

Step 1: Find the forward 90-day MRR 60-days from now.

[(1 + 0.054(150 / 360)) / (1 + 0.05(60 / 360)) − 1](360 / 90) = 0.056198. Since projected interest rates at the end of the FRA have increased to approximately 5.6%, which is above the contracted rate of 5%, the short position currently owes the long position.

Step 2: Find the interest differential between a loan at the projected forward rate and a loan at the forward contract rate.

(0.056198 − 0.05) × (90 / 360) = 0.0015495 × 10,000,000 = $15,495

Step 3: Find the present value of this amount 'payable' 90 days after contract expiration (or 60 + 90 = 150 days from now) and note once again that the short (who must 'deliver' the loan at the forward contract rate) loses because the forward 90-day MRR of 5.6198% is greater than the contract rate of 5%.

[15,495 / (1 + 0.054(150 / 360))] = $15,154.03

This is the *negative* value to the short.

(Module 30.4, LOS 30.c)

Question #21 of 77

Question ID: 1473661

The fixed-rate payer in an interest-rate swap has a position equivalent to a series of:

 rest-puts and short interest-rate calls.

B) long interest-rate puts and calls.

C) short interest-rate puts and long interest-rate calls.

Explanation

The fixed-rate payer has profits when short rates rise and losses when short rates fall, equivalent to writing puts and buying calls.

(Module 30.6, LOS 30.e)

Craig Champion, CFA, manages portfolios of U.S. securities for European investors. His clients have each hold different kinds of securities, and each has differing views with respect to hedging exchange rate risk.

Francois Levisque is a Belgian investor who holds a large diversified portfolio of U.S. equities. Levisque has a reputation for some success in timing the U.S. equity market. For example, he has often locked in gains on his portfolio with derivatives shortly before a market correction. Sometimes he also hedges his portfolio's currency risk.

Levisque has just instructed Champion to take a large short position in S&P 500 index, either with futures or with a forward contract. Champion notices that the futures price is less than the current spot price and consults with his colleague Danielle Silvers, CFA. Champion says he thinks that the futures price is less than the spot price because the dividend yield of the S&P 500 is greater than the Treasury Bill rate. Silvers says that it could just be backwardation.

Silvers also notes that the use of a forward contract might be a good idea because the contract will not attract the attention of other market participants who might react to Levisque's move. Champion tells Silvers that the reason Levisque wants to hedge his equity position is that he thinks all U.S. interest rates will increase soon. This, he believes, is bearish for equities.

Ragnar Hvammen is a Norwegian investor with a large investment in oil-related assets that he often hedges with futures contracts. Champion notices that the price of an oil futures contract is usually higher than the spot price. Hvammen uses short-term borrowings in dollars, from both European and U.S. banks, to meet the liquidity needs of his oil investments, and he has Champion hedge these loan positions with Eurodollar futures.

Silvers suggests that Champion should consider using T-bill futures to hedge the loans from U.S. banks, and use Eurodollar futures only for the Eurodollar loans. Champion says he will look into that, as well as forward rate agreements, as alternative hedging tools for Hvammen.

Champion is also evaluating pricing of Euro-bund futures. Specifically, he is looking for pricing on a 1.2-year contract. The CTD is a 2.5% 10-year, semi-annual coupon bond issued 1 year ago (just paid coupon) currently quoted at €104.10. The conversion factor for the bond is 1.08. At contract expiration, the underlying will have accrued interest of €0.42. Assume that the risk-free rate over the contract period is 1%.

Question #22 - 25 of 77

Question ID: 1586271

Champion and Silvers each gave a reason for why the futures price of the S&P 500 index might be less than the spot price. With respect to their statements, it is *most accurate* to conclude that:

A) neither statement is valid.

B) Champion's statement is invalid while Silver's statement is valid.

C) both statements are valid.

Explanation

The equation for the price of a futures contract on an equity index is $FP = S_0 \times e^{(R - \sigma) \times T}$, where σ is the dividend yield and R is the risk-free rate. If $R < \sigma$, then $FP < S_0$ and Champion is correct. Silvers could be correct in that backwardation is defined as $FP < S_0$, with the relationship being caused by the risk aversion of hedgers of long asset positions. Their risk aversion makes them willing to take short contracts at lower prices than otherwise might be the case.

(Module 30.3, LOS 30.d)

Question #23 - 25 of 77

Question ID: 1586272

For a futures contract on an asset with no storage costs, convenience yield, or other expected cash flows over the term of the contract, there should be a:

A) positive correlation between the futures price and interest rates and a negative correlation between the futures price and the spot price.

B) negative correlation between the futures price and interest rates and a positive correlation between the futures price and the spot price.

C) positive correlation between the futures price and both interest rates and the spot price.

The equation for the no-arbitrage price of a futures contract with no storage costs, convenience yield, or other expected cash flows over the term of the contract is $FP = S_0 \times (1 + R)^T$, so the futures price is positively correlated with both the interest rate and the spot price.

(Module 30.3, LOS 30.d)

Question #24 - 25 of 77

Question ID: 1586273

Oil futures prices might be higher than the spot price because:

A) of reverse contango.

B) there are more costs than benefits to holding the asset.

C) there are more benefits than costs to holding the asset.

Explanation

In calculating the futures price, we would subtract the benefits of holding the asset, e.g., the present value of dividends and coupons, and add the costs of holding the asset. Oil does not pay a dividend, and there would be costs for holding oil. Contango describes the situation where the futures price exceeds the spot price, and there is not such thing as reverse contango.

(Module 30.3, LOS 30.d)

Question #25 - 25 of 77

Question ID: 1586274

The no-arbitrage futures price of the Euro-bond contract is *closest* to:

A) €94.83.

B) €102.85.

C) €110.61.

Explanation

There will be 2 seminauunual coupon payments during the life of the futures contract: one in 6 months (t=0.5, T-t = 0.7) and one in one year (t=1, T-t = 0.2). Full price = €104.10 (since the bund just paid coupon, AI_0 = 0)

Step 1: compute the FV of 2 coupons: FVC = $(1.25 \times (1.01)^{0.7}) + (1.25 \times (1.01)^{0.2}) = 2.51$

Step 2: Compute Quoted Future price: QFP = [bund price × $(1 + R_f)^T$ − AI_T - FVC] /CF

$$= (104.10 \times (1.01)^{1.2} - 0.42 - 2.51)/1.08 = €94.83$$

(Module 30.3, LOS 30.d)

Question #26 of 77

Question ID: 1473656

A plain vanilla interest-rate swap to the fixed-rate payer is equivalent to issuing a fixed-rate bond and:

A) buying a floating-rate bond. ✓

B) selling a series of interest rate calls. ✗

C) selling a series of interest rate puts. ✗

Explanation

Paying fixed and receiving floating in a swap is equivalent to issuing a fixed-rate bond and investing the proceeds in a floating rate bond.

(Module 30.6, LOS 30.e)

Question #27 of 77

Question ID: 1508677

Consider a fixed-for-fixed 1-year $100,000 semiannual currency swap with rates of 5.0% in USD and 4.8% in CHF, originated when the exchange rate is $0.34. After the first settlement, the exchange rate is $0.35 and the term structure is:

	90 days	180 days
MRR	5.2%	5.6%
Swiss	4.8%	5.4%

What is the value of the swap to the USD payer?

A) [illegible] ✗

B) \$2,937.

C) -\$2,719.

Explanation

CHF periodic coupon (per 1 CHF) = 0.048/2 = 0.024

DF for 180 day CHF = 1 / (1 + 0.054 × (180/360)) = 1/1.027 = 0.9737

PV of CHF cash flows (per 1 CHF) = 0.9737 × 1.024 = 0.9971

At the current exchange rate the value is 0.9971 × 0.35 = USD 0.3490

The notional amount is 100,000/0.34 = 294,118 CHF so the dollar value of the CHF payments is 0.3490 × 294,118 = \$102,647.

USD periodic coupon (per 1 USD) = 0.05/2 = 0.025

DF for 180 day USD = 1 / (1 + 0.056 × (180/360)) = 1/1.028 = 0.9728

PV of USD cash flows (per 1 USD) = 0.9728 × 1.025 = 0.9971

Value (for notional = \$100,000) = 0.9971 × 100,000 = \$99,710.

The value of the swap to the dollar payer is 102,647 - \$99,710 = \$2,937.

(Module 30.7, LOS 30.f)

Question #28 of 77

Question ID: 1473658

For a 1-year quarterly-pay swap, an equivalent position with short puts and long calls would involve:

A) three put-call combinations expiring on the first three settlement dates of the swap.

B) put-call combinations expiring on each of the four settlement dates.

C) three put-call combinations on the last three settlement dates of the swap.

Explanation

Interest rate options pay one period after exercise. Options expiring on settlements at t = 1,2,3, will mimic the uncertain swap payments at t = 2,3,4.

(Module 30.6, LOS 30.e)

Question #29 of 77

Typesetting math: 100%

Question ID: 1473652

The price and value of a plain vanilla interest-rate swap are:

A) never equal.

B) only equal at the inception of a swap contract.

C) equal in equilibrium.

Explanation

The price of a swap is the fixed rate specified in the swap and is the same for the payer and the receiver. The value is the dollar value of the contract to the fixed-rate payer and is the opposite of the value to the floating-rate payer.

(Module 30.6, LOS 30.e)

Question #30 of 77

Question ID: 1473681

Consider a fixed-rate semiannual-pay equity swap where the equity payments are the total return on a $1 million portfolio and the following information:

- 180-day MRR is 5.2%
- 360-day MRR is 5.5%
- Dividend yield on the portfolio = 1.2%

What is the fixed rate on the swap?

A) 5.4197%.

B) 5.1387%.

C) 5.4234%.

Explanation

$$\frac{\left(1-\frac{1}{1.055}\right)}{\left(\frac{1}{1+0.052\left(\frac{180}{360}\right)}+\frac{1}{1+0.055\left(\frac{360}{360}\right)}\right)}=0.027117\times 2=5.4234\%$$

(Module 30.8, LOS 30.g)

Question #31 of 77

Question ID: 1473616

Typesetting math: 100% the value of a forward contract is:

A) always greater than or equal to zero.

B) the difference between the contract price and the market value of the underlying asset.

C) equal to the market price of the underlying asset.

Explanation

In a forward contract, the long is obligated to buy, and the short is obligated to sell, the underlying asset at the contract price. The difference between the contract price and the market price of the asset is what gives the contract value. The contract has a positive value at expiration to the long/short only if the contract price is below/above the market price.

(Module 30.1, LOS 30.b)

Question #32 of 77

Question ID: 1473624

The value of a futures contract between the times when the account is marked-to-market is:

A) never less than the value of a forward contract entered into on the same date.

B) equal to the difference between the price of a newly issued contract and the settle price at the most recent mark-to-market period.

C) the same as the contract price.

Explanation

Between the mark-to-market account adjustments, the contract value is calculated just like that of a forward contract; it is the difference between the price at the last mark-to-market and the current futures price, (i.e. the futures price on a newly issued contract). The mark-to-market of a futures contract is the payment or receipt of funds necessary to adjust for the gains or losses on the position. This adjusts the contract price to the 'no-arbitrage' price currently prevailing in the market.

(Module 30.1, LOS 30.b)

Elodie Brodeur works in the finance department of a large fashion house in France. The international catwalk season will start in two months' time and Brodeur has worked out that the company will need a 3-month loan of €4m in two months' time. The company's lenders are typically retail banks offering loans at MRR plus 40 basis points. Brodeur is concerned that interest rates may rise during the next two months and wants to use a FRA to lock-in the borrowing cost. Brodeur has collected the rate information shown in Exhibit 1.

Exhibit 1: Current MRR Rates

60 day MRR	2.0%
90 day MRR	2.4%
150 day MRR	2.6%
210 day MRR	2.9%

One month after the initiation of the FRA the MRR rates are shown in Exhibit 2.

Exhibit 2: MRR Rates One Month Later

30 day MRR	1.8%
90 day MRR	2.5%
120 day MRR	2.8%
180 day MRR	3.0%

At the expiration of the FRA, 90-day MRR is 3.4%.

Question #33 - 36 of 77

Question ID: 1473645

Which of the following comments relating to Brodeur's use of a forward rate agreement is *least* accurate?

A) A short FRA can be used to lock into a fixed rate of borrowing commencing in two months' time and expiring in five months' time.

B) The use of a FRA to hedge interest rate risk would lock Brodeur into paying a fixed rate plus 40 basis points for her borrowing.

C) The use of a FRA to hedge interest rate risk on her future loan will mean that she no longer benefits if interest rates fall.

Explanation

Typesetting math: 100%

A long FRA will create the obligation to pay fixed and receive floating from months 2 to month 5. The floating payments will offset her bank loan leaving her with a pay fixed obligation, where the fixed rate is determined today.

The floating receipt on the FRA and the floating payment on the loan she will need to take out will offset leaving a net payment of 40 basis points. Overall, she will now pay the fixed rate on the swap plus the 40 basis points on the loan.

The use of a long FRA will lock Elodie into paying fixed interest rate on the FRA (the FRAs forward price) plus the basis points on her loan above MRR. Elodie will no longer benefit from interest rate declines but will be protected from interest rises.

(Module 30.4, LOS 30.c)

Question #34 - 36 of 77

Question ID: 1473646

Using the data in Exhibit 1, which of the following is *closest* to the forward price of the FRA?

A) 2.4%.

B) 2.8%.

C) 3.0%.

Explanation

Step 1: Identify the correct MRR rates.

We will require the MRR rate until FRA expiry (day 60) and also the MRR rate at the end of the borrowing/lending period (day 150).

The 90-day MRR rate is a distractor. Elodie will borrow for 90 days but not from current date (T0). Elodie requires a 90-day loan commencing in 60 days' time.

Step 2: Unannualize the quoted rates.

MRR60 day = 2% × $\frac{60}{360}$ = 0.3333%

MRR150 day = 2.6% × $\frac{150}{360}$ = 1.0833%

Step 3: Compute the annualized forward price (fixed rate) starting in 60 days and lasting for 90 days.

Forward rate = ((1 + long rate) / (1 + short rate) – 1)(360 / 90) = 2.99%

(Module 30.4, LOS 30.c)

Question #35 - 36 of 77

Question ID: 1473647

For this question only, assume that the forward price of the FRA was 2.9%. Which of the following is the closest to the value accrued on the FRA from Brodeur's perspective one month after initiation of the contract?

A) +€2,300.

B) +€2,200.

C) –€2,265.

Explanation

The value of a FRA before expiry can be calculated by comparing the fixed rate on the original FRA to the fixed rate on a new FRA covering the same borrowing and lending period.

Step 1: Identify the correct MRR rates.

After 1 month has passed there is now 1 month until the original FRAs expiry and 4 months to the end of the borrowing and lending period. So we will require 30 day and 120 day MRR (i.e., (1 × 4) FRA price)

Step 2: Unannualize rates.

MRR30 day = 1.8% × $\frac{30}{360}$ = 0.15%

MRR120 day = 2.8% × $\frac{120}{360}$ = 0.9333%

Step 3: Compute the fixed rate (forward price) on a new FRA with the same expiry as the original FRA.

Forward rate = ((1 + long rate) / (1 + short rate) – 1)(360 / 90) = 3.1286%

Step 4: Compute the gain/(loss) on FRA at the end of the borrowing/lending period.

Gain to long = (new fixed rate – original fixed rate)(days / 360)(notional)

= (0.0313 – 0.029)(90 / 360)(4m) = 2,300

Step 5: Discount the gain/(loss) from the end of borrowing and lending to the valuation date (120 days).

$$\frac{€2{,}300}{1.009333} = €2,278$$

(Module 30.4, LOS 30.c)

Typesetting math: 100%

Question #36 - 36 of 77

Question ID: 1473648

Assuming the forward price of the FRA was 2.9%, which of the following is *closest* to the value of the FRA at expiration?

A) €4,958.

B) €5,000.

C) €19,831.

Explanation

Step 1: Compute gain/(loss) at end of borrowing and lending period.

$$(\text{floating rate} - \text{fixed rate})\left(\frac{90}{360}\right)(\text{notional principal})$$

$$(0.034 - 0.029)\left(\frac{90}{360}\right)(€4\text{m}) = €5,000$$

Step 2: Discount the gain/(loss) back to FRA expiry (90 days).

MRR90 day = 3.4% × $\frac{90}{360}$ = 0.85 %

$$\frac{€5{,}000}{1.0085} = €4,958$$

(Module 30.4, LOS 30.c)

Question #37 of 77

Question ID: 1473674

In an equity return swap for MRR, if the return on the underlying equity portfolio is negative for a payment period, the equity return payer will:

A) receive a net payment greater than the loss of value on the equity portfolio.

B) make a net payment greater than the loss of value on the equity portfolio.

C) receive a net payment less than the loss of value on the equity portfolio.

Explanation

The equity return payer will receive the periodic interest payment and "pay" the negative return on the portfolio, resulting in a net payment to the equity return payer that is greater than the loss on the equity portfolio.

(Module 30.8, LOS 30.g)

Question #38 of 77

Question ID: 1473632

A stock is currently priced at \$110 and will pay a \$2 dividend in 85 days and is expected to pay a \$2.20 dividend in 176 days. The no arbitrage price of a six-month (182-day) forward contract when the effective annual interest rate is 8% is *closest* to:

A) \$110.00.

B) \$110.06.

C) \$110.20.

Explanation

In the formulation below, the present value of the dividends is subtracted from the spot price, and then the future value of this amount at the expiration date is calculated.

$$(110 - 2/1.08^{85/365} - 2.20/1.08^{176/365})\ 1.08^{182/365} = \$110.06$$

Alternatively, the future value of the dividends could be subtracted from the future value of the stock price based on the risk-free rate over the contract term.

(Module 30.2, LOS 30.a)

Question #39 of 77

Question ID: 1473635

Consider a 9-month forward contract on a 10-year 7% Treasury note just issued at par. The effective annual risk-free rate is 5% over the near term and the first coupon is to be paid in 182 days. The price of the forward is *closest* to:

A) 965.84.

B) 1,037.27.

C) 1,001.84.

Explanation

The forward price is calculated as the bond price minus the present value of the coupon, times one plus the risk-free rate for the term of the forward.

$$(1{,}000 - 35/1.05^{182/365})\ 1.05^{9/12} = \$1{,}001.84$$

(Module 30.3, LOS 30.d)

Typesetting math: 100%

Question #40 of 77

Question ID: 1473683

Consider a 1-year, $5 million semiannual-pay fixed-rate equity swap initiated when the equity index is 750 and swap fixed rate is 3.7%. Equity index was at 760 at first settlement. It is now 270 days since inception of the swap and the index is at 767, 90-day MRR is 3.4% (DF = 0.99157) and 270-day MRR is 3.7% (DF = 0.9730). What is the value of the swap to the fixed-rate payer?

A) $3,478.

B) −$2,726.

C) −$3,520.

Explanation

For $100 notional, the value of the equity side is (767/760) x $100 = $100.921

Value of the semiannual –pay, fixed rate bond with 3.7% annual coupon =[100 + 3.7/2] x 0.99157 = $100.9914

Value of pay-fixed side = value of equity – value of fixed rate bond = $100.921 - $100.9914 = -$0.0704 (per $100 notional).

For $5 million notional, value = 50,000 x -0.0704 = -$3,520

(Module 30.8, LOS 30.g)

Question #41 of 77

Question ID: 1473629

The value of the S&P 500 Index is 1,260. The continuously compounded risk-free rate is 5.4% and the continuous dividend yield is 3.5%. Calculate the no-arbitrage price of a 160-day forward contract on the index.

A) $562.91.

B) $1,270.54.

C) $1,310.13.

Explanation

$FP = 1{,}260 \times e^{(0.054 - 0.035) \times (160 / 365)} = 1{,}270.54$

(Module 30.2, LOS 30.a)

Question #42 of 77

Question ID: 1586275

What is the value of a 6.00% 1x4 (30 days x 120 days) forward rate agreement (FRA) with a principal amount of $2,000,000, 10 days after initiation if the MRR curve at that time reflects the following:

110-day MRR = 6.15% and

20-day MRR = 6.05%?

A) $745.76.

B) $767.40.

C) $700.00.

Explanation

The current 90-day forward rate at the settlement date, 20 days from now is:

([1+ (0.0615 x 110/360)]/[1+ (0.0605 x 20/360)] – 1) x 360/90 = 0.061517

The interest difference on a $2 million, 90-day loan made 20 days from now at the above rate compared to the FRA rate of 6.0% is:

[(0.061517 x 90/360) – (0.060 x 90/360)] x 2,000,000 = $758.50

Discount this amount at the current 110-day rate:

758.50/[1+ (0.0615 x 110/360)] = $745.76

(Module 30.4, LOS 30.c)

Question #43 of 77

Question ID: 1473620

The theoretical price of a forward contract:

A) equals the long's expectation of the future price of the underlying asset.

B) is the no-arbitrage price.

C) is always greater than the current price of the underlying asset.

Explanation

The theoretical price of a forward contract is the future price of the underlying asset imposed by the no-arbitrage conditions. It can be less than the current price of the asset if the cost-of-carry is negative. Accrued interest is paid by the long at delivery under a bond forward, but is not included in the price quote, which is usually in terms of yield to maturity at the settlement date.

(Module 30.1, LOS 30.b)

Question #44 of 77

Question ID: 1473668

A U.S. firm (U.S.) and a foreign firm (F) engage in a 3-year, annual pay currency swap; The USD fixed rate at initiation was 5% while FC fixed rate was 4%. At the beginning of the swap, $2 million was paid by the U.S. firm and the exchange rate was 2 FC units per $1. At the end of the swap period the exchange rate was 1.75 FC units per $1.

At the end of year 1, firm:

A) U.S. pays firm F 160,000 FC units. ✓

B) F pays firm U.S. $200,000. ✗

C) U.S. pays firm F $200,000. ✗

Explanation

Firm U.S. pays fixed 4% on FC = 0.04 × $2,000,000 × 2 FC units per $1 = 160,000 FC units.

(Module 30.7, LOS 30.f)

Question #45 of 77

Question ID: 1473663

Which of the following is *equivalent* to a pay-fixed swap with a tenor of two years with semi-annual swap payments and a fixed rate of 6% (exchanged for MRR)? The notional principal is $100,000,000.

A) A strip of three forward rate agreements, which obligates the party to pay a fixed rate of 6% and receive six-month MRR on a notional principal of $100,000,000. ✓

B) A forward rate agreement, which obligates the party to pay a fixed rate of 6% and receive six-month MRR on a notional principal of $100,000,000. ✗

C) A strip of two forward rate agreements, which obligates the party to pay a fixed rate of 6% and receive six-month MRR on a notional principal of $100,000,000. ✗

Explanation

In an interest rate swap, the first payment is known with certainty and will be made at month 6. The determination dates for the floating rate will be at months 6, 12, and 18 and the corresponding payment dates will be at months 12, 18, and 24. These correspond to the three forward rate agreements.

(Module 30.6, LOS 30.e)

Question #46 of 77

Question ID: 1473634

Calculate the price of a 200-day forward contract on an 8%, semi-annual, U.S. Treasury bond with a spot price of $1,310. Next coupon payment will be made in 150 days. The annual risk-free rate is 5%.

A) $1,333.50.

B) $1,270.79.

C) $1,305.22.

Explanation

Coupon = (1,000 × 0.08) / 2 = $40.00

Present value of coupon payment = $40.00 / $1.05^{150/365}$ = $39.21

Forward price on the fixed income security = ($1,310 - $39.21) × $(1.05)^{200/365}$ = $1,305.22

(Module 30.3, LOS 30.d)

Question #47 of 77

Question ID: 1473615

The price of a forward contract:

A) depends on forward interest rates.

B) is determined at contract initiation.

C) changes over the term of the contract.

Explanation

The price of a forward contract is established at the initiation of the contract and is expressed in different terms, depending on the underlying assets. It is the price that makes the contract value zero, and depends on current interest rates through the cost-of-carry calculation.

(Module 30.1, LOS 30.b)

Question #48 of 77

Question ID: 1473667

Which of the following is equivalent to a pay-floating USD receive-fixed EUR currency swap position?

Typesetting math: 100%

A) A short position in a EUR bond coupled with the issuance of a USD-denominated floating rate note.

B) A long position in a EUR bond coupled with the issuance of a USD-denominated floating rate note.

C) A short position in a EUR bond coupled with a long position in a USD-denominated floating rate note.

Explanation

A long position in a fixed rate EUR bond will receive fixed coupons denominated in EUR. The short floating rate note requires USD denominated floating-rate payments. Combined, these are the same cash flow as a pay-floating USD receive-fixed EUR currency swap.

(Module 30.7, LOS 30.f)

Question #49 of 77

Question ID: 1473660

Which of the following is *equivalent* to a plain vanilla receive-fixed interest rate swap?

A) A short position in a bond coupled with the issuance of a floating rate note.

B) A short position in a bond coupled with a long position in a floating rate note.

C) A long position in a bond coupled with the issuance of a floating rate note.

Explanation

A long position in a fixed rate bond receives fixed coupons. The short floating rate note requires floating-rate payments. Together, these are the same cash flow as a receive-fixed swap.

(Module 30.6, LOS 30.e)

Question #50 of 77

Question ID: 1473628

Over the life of a swap, the price of the swap:

A) is approximately equal to the market value of the swap.

B) fluctuates with changes in the yield curve.

C) does not change.

Explanation

The price of a swap, quoted as the fixed rate in the swap, is determined at contract initiation and remains fixed for the life of the swap.

(Module 30.1, LOS 30.b)

Question #51 of 77

Question ID: 1587862

The fixed-rate receiver in a plain vanilla interest rate swap has a position equivalent to a series of:

A) short interest-puts and long interest-rate calls.

B) long interest-rate puts and short interest-rate calls.

C) long interest-rate puts.

Explanation

The fixed-rate receiver has profits when short rates fall and losses when short rates rise, equivalent to buying puts and writing calls.

(Module 31.6, LOS 31.j)

Question #52 of 77

Question ID: 1473684

Consider a 1-year semiannual equity swap based on an index at 985 and a fixed rate of 4.4%. 90 days after the initiation of the swap, the index is at 982 and MRR is 4.6% for 90 days and 4.8% for 270 days. The value of the swap to the equity payer, based on a $2 million notional value is *closest* to:

A) −$22,564

B) $22,314

C) $22,564

Explanation

$$= \frac{982}{985} - \frac{\frac{0.044}{2}}{1+\left(0.046\times\frac{90}{360}\right)} - \frac{\frac{0.044}{2}}{1+\left(0.048\times\frac{270}{360}\right)} - \frac{1}{1+\left(0.048\times\frac{270}{360}\right)}$$

$$= \frac{982}{985} - \frac{0.022}{1.0115} - \frac{0.022}{1.036} - \frac{1}{1.036}$$

$$= 0.996954 - 0.021750 - 0.021236 - 0.965251$$

$$= -0.0112821 \times 2,000,000 = -\$22,564$$

−$22,564 is the value to the fixed-rate payer, thus $22,564 is the value to the equity return payer.

(Module 30.8, LOS 30.g)

Question #53 of 77

Question ID: 1473621

At contract initiation, the value of a forward contract:

A) is typically zero regardless of the price of the underlying asset. ✓

B) is set to 100 by convention. ✗

C) depends on the market price of the underlying asset. ✗

Explanation

Due to the no-arbitrage principle, the price of a forward contract is calculated to make the value of the contract zero at contract initiation. Neither the long nor the short typically makes any payment to enter into the forward agreement. A special case is an off-market forward where, for whatever reason, the contract price is not set equal to the no-arbitrage price, and the long or short position makes a payment to the opposite counterparty to offset the difference.

(Module 30.1, LOS 30.b)

Question #54 of 77

Question ID: 1473619

The price of a forward contract:

A) is equal to the value of the contract in equilibrium. ✗

B) is the settlement price for the underlying asset. ✓

 equal to the market price at contract termination. ✗

Explanation

The price of a forward contract is the price of the underlying asset that the long will pay to the short at settlement (for a deliverable contract). The value of a forward contract comes from the difference between the forward contract price and the market price for the underlying asset. This difference between price and value is a key concept to understand. A forward contract has only one price, which applies to both the long and to the short.

(Module 30.1, LOS 30.b)

Abel Smith works in the Treasury Department of OTS Ltd. OTS is an international construction firm, based in the United States. OTS hopes to raise €100 million through the issuance of a €100 million one-year fixed rate bond but is concerned about the currency risk exposure.

TNA Bank proposed a one year EUR-USD currency swap with semi-annual settlements to OTS to mitigate the exchange rate risk. The notional principal would be €100 million. The bank provides the following information:

Exhibit 1: MRR Spot Rate (annualized)

Days	$ MRR	PV Factors	Days	MRR	PV Factors
180	0.70%	0.9965	180	1.50%	0.9926
360	1.00%	0.9901	360	2.00%	0.9804

Using the information in Exhibit 1, the bank calculates that the currency swap's fixed rates are 0.85% on the USD and 1.75% on the euro.

The term structure of MRR and MRR spot rates three months after the swap initiation is shown in Exhibit 2:

Exhibit 2: MRR Spot Rate (annualized)

Days	$ MRR	PV Factors	Days	MRR	PV Factors
90	0.85%	0.9979	90	1.30%	0.9968
270	1.20%	0.9911	270	2.30%	0.9830

Exhibit 3: Exchange Rates

Time	Exchange Rate	Time	Exchange Rate
Swap initiation	€1: USD1.43	1st settlement date	€1: USD1.55
90 days after swap initiation	€1: USD1.48	2nd settlement date	€1: USD1.55

OTS has an investment portfolio with similar weighting as the S&P500. Smith believes that the U.S. equity market could suffer further declines and OTS could hedge the equity risk using an equity swap. Smith obtained the outlook of the U.S. equity market in Exhibit 4.

Exhibit 4: Performance of S&P500 for the Next Three Quarters

Time	S&P Index Level
Swap initiation	1,190
1st quarter	1,000
2nd quarter	980
3rd quarter	1,050
4th quarter	1,130

Smith is proposing for OTS to be the party paying the equity return on a USD500 million one-year equity swap. OTS will be receiving fixed rate of 1% on a semi-annual basis.

Smith makes two comments:

Comment 1:	A fixed-rate payer in an equity swap would not have to pay more than the fixed rate each period.
Comment 2:	Compared to an interest rate swap, the first payment in an equity swap will not be known at initiation.

Typesetting math: 100%

Question #55 - 58 of 77

Question ID: 1473677

Has the bank correctly calculated the fixed rates on the currency swap?

A) Both fixed rates are incorrectly calculated.

B) One of the two fixed rates is incorrectly calculated.

C) Both fixed rates are correctly calculated.

Explanation

Fixed rate for the USD:

$$\frac{1-0.9901}{0.9965+0.9901} = 0.00498 = 0.498\%$$

In annual terms, the fixed rate for the USD = $0.498\% \times \frac{360}{180} = 0.997\%$

Fixed rate for the euro:

$$\frac{1-0.9804}{0.9926+0.9804} = 0.00993 = 0.993\%$$

In annual terms, the fixed rate for the euro = $0.993\% \times \frac{360}{180} = 1.987\%$

The fixed rates quoted by the bank are both incorrectly calculated.

(Module 30.7, LOS 30.f)

Question #56 - 58 of 77

Question ID: 1496747

If OTS decides on a fixed for fixed currency swap, what is the market value of the swap to OTS three months after swap initiation?

A) The value of the euro payments is €97.68m and converting the euro value to USD using €1: USD1.55, the swap has a positive to OTS.

B) The value of the euro payments is €100.27m and converting the euro value to USD using €1: USD1.48, the swap has a positive to OTS.

C) The value of the euro payments is €101.8m and converting the euro value to USD using €1: USD1.43, the swap has a positive to OTS.

Explanation

Typesetting math: 100%

To mitigate currency risk, OTS would be the party paying the fixed rate on USD and receiving the fixed rate on euro. The notional principal is €100m to USD143 million. Note this is using the $/€ of 1.43 at swap initiation.

We show the computation of the value of USD payments also even though it is not required for the question. The value of the USD fixed payments is equivalent to the value of a USD143m fixed rate bond that pays a coupon in 90 days' time and a coupon plus face value in 270 days' time. The coupon rate *would* be 0.85% as provided in the vignette.

The coupons on the bond = 0.0085 × (180 / 360) × $143m = $607,750.

Maturity	$ MRR	Discount Factor	CF	DCF
90	0.85%	0.9979	$607,750	606,474
270	1.20%	0.9911	$143,607,750	142,329,641
			PV $ Fixed	$142,936,115

The value of the Euro fixed payments is equivalent to the value of a €100m fixed rate bond that pays a coupon in 90 days' time and a coupon plus face value in 270 days' time. The coupon rate would be 1.75% as computed in the solution to question 1.

The coupons on the bond = 0.0175 × (180 / 360) × €100m = €875,000

Maturity	$ MRR	Discount Factor	CF	DCF
90	1.30%	0.9968	€875,000	872,200
270	2.30%	0.9830	€100,875,000	99,160,125
			PV € Fixed	€100,032,325

OTS is based in the U.S. but has borrowed in €'s. OTS would therefore take the position of receiving €'s and paying $'s in the currency swap.

Convert the € fixed coupon bond into $'s at the date of the valuation using the exchange rate $/€ = 1.48

€100,032,325 × 1.48 = $148,047,841

The value to the party paying $'s and receiving €'s = $148,047,841 – $142,936,115 = $5,111,726

Note that no actual computations were required to answer this question if you spotted that answer B was the only solution to use the correct exchange rate.

Typesetting math: 100%

(Module 30.7, LOS 30.f)

Question #57 - 58 of 77

Question ID: 1473679

Using the information in Exhibit 2, what is the market value of the equity swap to OTS three months after swap initiation?

A) +USD80.35 million.

B) +USD85.33 million.

C) +USD88.76 million.

Explanation

OTS is paying the equity return and the value is:

(1000 / 1190) × USD500m = USD420.17 million

The value of fixed payments is equivalent to the value of a one-year fixed coupon bond with 0.5% semi-annual coupon. The value of bond is the present value of the two coupon payments and the par value:

[(0.005 × 0.9979) + (1.005 × 0.9911)] × 500m = USD 500.52 million.

Value to OTS: USD500.52m – USD420.17m = USD80.35 million

(Module 30.8, LOS 30.g)

Question #58 - 58 of 77

Question ID: 1473680

How many of Smith's comments are correct?

A) Neither comment is correct.

B) One comment is correct.

C) Both comments are correct.

Explanation

When the index declines, the fixed rate payer would pay the negative return in addition to the fixed rate and, hence, will suffer a loss greater than the fixed rate. There is no way of knowing the first payment on an equity swap because we do not know the value of the equity index on the payment date. The floating rate for the first settlement also known at time 0 (known as advance set, paid in arrears).

(Module 30.8, LOS 30.g)

Question #59 of 77

Question ID: 1473666

A $10 million 2-year semi-annual-pay MRR-based interest-rate swap was initiated 180 days ago when swap fixed rate was 3.8%. The fixed rate on the swap is now 3.4% and the term structure is as follows:

Days	MRR	Discount Factor
180	3.00%	0.98522
360	3.20%	0.96899
540	3.40%	0.95148
720	4.00%	0.92593

Value of the swap to the payer is *closest* to:

A) -$45,633.

B) -$58,114.

C) -$29,229.

Explanation

Sum of the discount factors for the three settlement dates remaining (180,360 and 540 days away) = 0.98522 + 0.96899 + 0.95148 = 2.9057

Value to payer = 2.9057 x [(0.034 – 0.038)/2] x $ 10 million = -$58,114

(Module 30.6, LOS 30.e)

Question #60 of 77

Question ID: 1473653

Typesetting math: 100% n interest rate swap is the:

A) cost to purchase a swap.

B) fixed rate of interest.

C) market value of the swap.

Explanation

The price of an interest rate swap is quoted as the rate on the fixed-rate payments. The floating rate is a known reference rate but does not need to be quoted.

(Module 30.6, LOS 30.e)

Question #61 of 77

Question ID: 1473670

In which type of swap contract is notional principal *most likely* to be exchanged at initiation?

A) Currency swap.

B) Equity swap.

C) Interest rate swap.

Explanation

Notional principal is typically exchanged at initiation of a currency swap, but not in an interest rate swap or equity swap.

(Module 30.7, LOS 30.f)

Question #62 of 77

Question ID: 1473675

An equity portfolio manager who has a positive long-term outlook for equities, but expects equity prices to decline over the next three months, would *most appropriately* enter into:

A) a short straddle with an equity index as the underlying.

B) an equity swap as the equity return receiver.

C) an equity swap as the equity return payer.

Explanation

By entering an equity swap as the equity return payer, the manager can protect the portfolio value from a short-term decline in equity prices while keeping ownership of the equities for the longer term. A short straddle costs the investor when the price of the underlying moves up or down.

(Module 30.8, LOS 30.g)

Question #63 of 77

Question ID: 1473623

What is the difference between spot and futures prices? Spot prices are always:

A) delivered to meet the futures obligation at expiration.

B) equal to the futures price at futures expiration.

C) lower than futures prices.

Explanation

The difference between the spot and the futures price must be zero at expiration to avoid arbitrage.

(Module 30.1, LOS 30.b)

Question #64 of 77

Question ID: 1473626

During the life of a forward contract, the value of the contract is *best* described as:

A) the difference between the spot price and the present value of the forward price of the underlying asset.

B) the difference between the future value of the spot price and the expected future price of the underlying asset.

C) the present value of the expected future price of the underlying asset.

Explanation

The value of a forward contract on an asset with no cash flows during its term is equal to spot − (forward price) / $(1 + R_f)^t$), the difference between the spot price and the present value of the forward price of the underlying asset.

(Module 30.1, LOS 30.b)

Question #65 of 77

Question ID: 1586282

The floating-rate payer in a simple interest-rate swap has a position that is equivalent to:

A) a series of long forward rate agreements (FRAs).

Typesetting math: 100%

B) a series of short FRAs.

C) issuing a floating-rate bond and a series of long FRAs.

Explanation

The floating-rate payer has a liability/gain when rates increase/decrease above the fixed contract rate; the short position in an FRA has a liability/gain when rates increase/decrease above the contract rate.

(Module 30.4, LOS 30.c)

Question #66 of 77

Question ID: 1473682

Consider a fixed-rate semiannual-pay equity swap where the equity payments are the total return on a $1 million portfolio and the following information:

- 180-day MRR is 4.2%
- 360-day MRR is 4.5%
- Div. yield on the portfolio = 1.2%

What is the fixed rate on the swap?

A) 4.3232%.

B) 4.5143%.

C) 4.4477%.

Explanation

$$\frac{\left(1-\frac{1}{1.045}\right)}{\left(\frac{1}{1+0.042\left(\frac{180}{360}\right)}+\frac{1}{1+0.045\left(\frac{360}{360}\right)}\right)}=0.022239\times 2=4.4477\ \%$$

(Module 30.8, LOS 30.g)

Question #67 of 77

Question ID: 1473643

A company has chosen to use a 6 x 9 FRA expiring in 6 months to mitigate the risk of paying a floating coupon on the bond issue. The current term structure for MRR is as follows:

Term	Interest Rate
180 days	5.65%
270 days	5.95%

What is the price of this forward rate agreement (FRA)?

A) 3.19%

B) 6.37%

C) $6.37

Explanation

The price of an FRA is the fixed rate. To determine the FRA's fixed rate, the following formula should be used:

$$\text{FRA price} = \left(\frac{P_{Y_{1+r}}}{P_{X_{1+r}}} - 1\right)\left(\frac{360}{Y-X}\right)$$

$$= \left[\frac{1+.0595\left(\frac{270}{360}\right)}{1+.0565\left(\frac{180}{360}\right)} - 1\right]\left(\frac{360}{90}\right) = \underline{\underline{0.0637}}$$

The FRA's fixed rate would be quoted as 6.37%.

The price of an FRA is given as a rate percentage, never as a dollar amount.

(Module 30.4, LOS 30.c)

Question #68 of 77

Question ID: 1473631

Calculate the no-arbitrage forward price for a 90-day forward on a stock that is currently priced at $50.00 and is expected to pay a dividend of $0.50 in 30 days and a $0.60 in 75 days. The annual risk free rate is 5% and the yield curve is flat.

A) $48.51.

B) $50.31.

C) $49.49.

Explanation

Typesetting math: 100%

The present value of expected dividends is: \$0.50 / ($1.05^{30/365}$) + \$0.60 / ($1.05^{75/365}$) = \$1.092

Future price = (\$50.00 − 1.092) × $1.05^{90/365}$ = \$49.49

(Module 30.2, LOS 30.a)

Question #69 of 77

Question ID: 1473642

The price of a 3 × 5 forward rate agreement (FRA) is the:

A) 2-month implied forward rate 3 months from today. ✓

B) 3-month implied forward rate 5 months from today. ✗

C) 2-month implied forward rate 5 months from today. ✗

Explanation

The notation for FRAs is unique. There are two numbers associated with an FRA: the number of months until the contract expires and the number of months until the underlying loan is settled. The difference between these two is the maturity of the underlying loan. For example, a 3 × 5 FRA is a contract that expires in three months (90 days), and the underlying loan is settled in five months (150 days). The price of the 3 × 5 FRA is calculated by annualizing the implied forward rate. The implied forward rate is calculated from the 3-month rate and the 5-month rate.

(Module 30.4, LOS 30.c)

Question #70 of 77

Question ID: 1473657

A swap is equivalent to a series of:

A) FRAs priced at market rates. ✗

B) off-market FRAs. ✓

C) interest rate calls. ✗

Explanation

Since the fixed rate on the swap is the same at every settlement date, a series of FRAs at those fixed rates will have values that differ from zero to the extent the fixed rate and the zero-value rate differ. This makes them off-market FRAs.

(Module 30.6, LOS 30.e)

John Williams, CFA, works in the treasury department of Sam Smith Leisure Inc., a U.S. based manufacturer of gym equipment. Recently he has been considering using derivative instruments to lock in returns on excess cash flows that tend to accumulate in the final quarter of each year as demand for equipment peaks during that time.

He estimates that this year, in 60-days, the company will have $28.5 million in excess funds to invest for 90 days.

Williams is presenting to the board 60 days before the excess funds need to be deposited, which is also 30 days before the year end. He intends to suggest an FRA as a method of locking in a return on the deposit. He intends to make the following two statements in favor of using an FRA.

Statement 1

As we are depositing cash, committing to an FRA will generate a cash inflow on the date we enter into it.

Statement 2

If rates move in our favor, we will receive a cash payment at the end of the notional borrowing period.

Williams will present the hypothetical rates and MRR shown in Exhibit 1 to illustrate the result of using an FRA. Current 2x5 FRA price is 3.8%. All rates are annualized.

Exhibit 1 – FRA Price and Theoretical Future MRR rates

Predicted MRR rates	In 30 days	In 60 days	In 90 days	In 120 days	In 150 days
30-day MRR	3.9%	4.0%	4.2%	4.4%	4.5%
60-day MRR	4.1%	4.4%	4.5%	4.7%	4.8%
90-day MRR	4.2%	4.7%	4.8%	4.9%	5.2%
120-day MRR	4.5%	5.0%	5.2%	5.3%	5.5%
150-day MRR	4.8%	5.3%	5.4%	5.6%	5.9%

One key question that the CFO is likely to ask is the predicted value of the FRA at the year end.

Finally Williams is to investigate the potential for Sam Smith Inc. to use a currency swap to borrow and invest in a manufacturing facility in Europe. A bank has offered the company a fixed for fixed currency swap involving US dollars and Euros.

Some of the swap details are outlined in Exhibit 2.

Exhibit 2– Currency Swap

Initiation:	1st January
Spot rate at initiation:	USD/EUR 1.19
Settlement:	Quarterly
Principal:	USD 40,000,000
Fixed EUR rate:	1.5%
Fixed USD rate:	1.3%

Question #71 - 74 of 77

Question ID: 1586284

Which of Williams' statements regarding FRAs is *most likely* correct?

A) Only Statement 2 is correct.

B) Neither statement is correct.

C) Only Statement 1 is correct.

Explanation

An FRA involves no initial cash flow as it is a commitment. The FRA will pay off at the date of expiry (the start of the notional borrowing period).

(Module 30.7, LOS 30.f)

Question #72 - 74 of 77

Question ID: 1586285

Using the price and predicted MRR rates in Exhibit 1, which of the following is closest to the predicted value of the FRA at the year end?

A) –$100,000

B) –-$85,000

C) –$62,000

Explanation

Typesetting math: 100%

The year end is 30 days away. At that point the required FRA would be a 1X4 (90 days borrowing in 30 days' time).

To value the FRA, first price this FRA to compare to the current FRA price.

30 day MRR at year end 3.9% × 30/360 = 0.325%

120 day MRR at year end 4.5% × 120/360 = 1.5%

FRA price = 1.015/1.00325 – 1 = 0.01171

Quoted annually = 0.01171 × 360/90 = 0.046848 = **4.685%**

Value = (3.8% – 4.685%) × $28.5 million × 90/360 = –$63,056

Discounted to year end = –$63,056/1.015 = **–$62,124**

(Module 30.7, LOS 30.f)

Question #73 - 74 of 77

Question ID: 1586286

Using the details shown in Exhibit 2, under the terms of the currency swap at the first settlement date Sam Smith would *most likely*:

A) pay EUR 150,000.

B) pay USD 130,000.

C) pay EUR 126,050.

Explanation

Sam Smith is using the currency swap to obtain EUR funding. It will therefore receive EUR (and pay USD) at the outset of the swap and pay EUR interest (and receive USD) at each settlement date.

The USD notional principal is 40,000,000. At a spot rate of 1.19 that equates to EUR 33,613,445

The interest payment is therefore 33,613,445 × 1.5% × 90/360 = **126,050**

(Module 30.7, LOS 30.f)

Question #74 - 74 of 77

Question ID: 1586287

Which of the following statements regarding cash flows at the final settlement date for the currency swap outlined in Exhibit 2is *most likely* correct?

A) Without knowing the spot rates on the final settlement date, it is impossible to state the cash flows that occur.

B) Sam Smith Inc. will receive USD 40,000,000 plus the USD interest payment.

C) Sam Smith will pay USD 40,000,000 and receive the final USD interest payment.

Explanation

Smith initially borrowed EUR and swapped USD. Throughout the swap, at each settlement date Smith will pay EUR interest and receive USD.

At the final settlement date, Smith will receive the final USD interest payment and receive the USD principal back.

(Module 30.7, LOS 30.f)

Question #75 of 77

Question ID: 1473630

An index is currently 965 and the continuously compounded dividend yield on the index is 2.3%. What is the no-arbitrage price on a one-year index forward contract if the continuously compounded risk-free rate is 5%.

A) 991.1.

B) 987.2.

C) 991.4.

Explanation

The futures price $FP = S_0 e^{-\delta T} \, (e^{RT})$

$= S_0 e^{(R-\delta)T}$

$= 965 e^{(.05-.023)}$

$= 991.4$

(Module 30.2, LOS 30.a)

Question #76 of 77

Question ID: 1473633

An index is currently 876, the risk-free rate (R_f) is 7%, and the dividend yield on the index portfolio is 1.8%. Assuming that these are continuously compounded yields, the price of an 18-month index future is *closest* to:

Typesetting math: 100%

A) 947.1.

B) 943.0.

C) 945.2.

Explanation

FP = 876 $e^{(0.07-0.018)1.5}$ = 947.1.

(Module 30.2, LOS 30.a)

Question #77 of 77

Question ID: 1473618

Which of the following *best* describes the price of a forward contract? The forward price is:

A) always equal to the market price at contract termination.

B) always expressed in dollars.

C) the price that makes the values of the long and short positions zero at contract initiation.

Explanation

The forward price is the contract price of the underlying asset under the terms of the forward contract, and is the price that makes the values of the long and short positions zero at contract initiation. It is not the amount it costs to purchase the forward contract. The forward price is expressed in terms of the underlying asset, and may be a dollar value, exchange rate, or interest rate. The value of a forward contract comes from the difference between the forward contract price and the market price for the underlying asset. These values are likely to be different at contract termination, which will result in a profit for either the long or the short position.

(Module 30.1, LOS 30.b)